॥वसुधैव कुटुम्बकम्॥

SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Symbiosis International (Deemed University)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Assignment No. 02

Subject:	DevOps Lab
TCODE:	TE7950
Name of Student	**Raghav Sharma**
PRN No.	**23070122171**
Branch	CSE, Batch (2023-27), L3
Academic Year & Semester	25 – 26, Sem VII
Date of Performance	23rd July, 2026
Title of Assignment:	Git Collaboration & Conflict Resolution
Theory:	Git branching allows developers to work on different features independently without affecting the main project. Branches can be merged after successful development and testing. This approach supports parallel development, simplifies collaboration, and helps maintain code stability by integrating changes in a controlled manner.

Implementation:	Step 1: Create a new branch named conflict and verify the available branches using the git branch command.
	`C:\Users\sharm\Desktop\7th sem\DevOpsL\devopsL\Github>git branch conflict`
	
	`C:\Users\sharm\Desktop\7th sem\DevOpsL\devopsL\Github>git branch`
	`  conflict`
	`  feature`
	Step 2: Modify the README.md file in the main branch, stage the changes using git add ., and commit them using git commit -m "Making Conflict".
	`C:\Users\sharm\Desktop\7th sem\DevOpsL\devopsL\Github>git add .`
	
	`C:\Users\sharm\Desktop\7th sem\DevOpsL\devopsL\Github>git commit -m "Making Conflict"`
	`[main c57ec4d] Making Conflict`
	`1 file changed, 1 insertion(+), 1 deletion(-)`

	Step 3: Switch to the conflict branch using git checkout conflict, modify the same file, stage the changes, and commit them using git commit -m "On Conflict".
	`:\Users\sharm\Desktop\7th sem\DevOpsL\devopsL\Github>git checkout conflict`
	`witched to branch 'conflict'`
	
	`:\Users\sharm\Desktop\7th sem\DevOpsL\devopsL\Github>git add .`
	Step 4: Switch back to the main branch using git checkout main.
	`C:\Users\sharm\Desktop\7th sem\DevOpsL\devopsL\Github>git checkout main`
	`Switched to branch 'main'`
	Step 5: Merge the conflict branch into the main branch using git merge conflict and observe the merge conflict message.
	`C:\Users\sharm\Desktop\7th sem\DevOpsL\devopsL\Github>git merge conflict`
	`Auto-merging readme.md`
	`CONFLICT (content): Merge conflict in readme.md`
	`Automatic merge failed; fix conflicts and then commit the result.`

Subject Faculty Name: Prof. Arijit Dutta